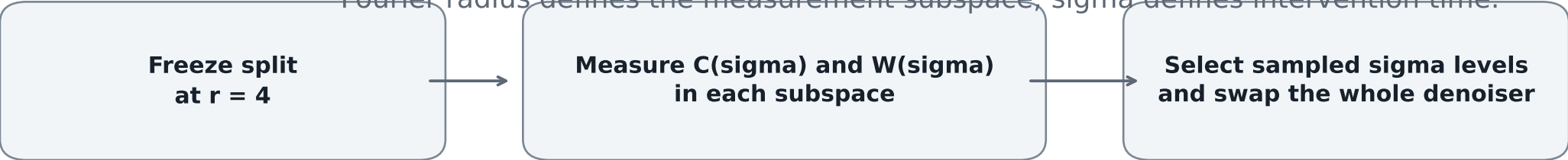


Frequency geometry selects when to swap the denoiser

Fourier radius defines the measurement subspace; sigma defines intervention time.



LOW FREQUENCY

$r \leq 4$

Geometry: $C_{\text{low}}(\sigma)$, $W_{\text{low}}(\sigma)$

selects one sampled candidate at $\sigma = 3.257$

Condition	Noise level (sigma)	Final memorization
Before control	5.315	80.08%
Candidate	3.257	73.44%
After control	1.923	71.09%

Result: low-derived suppression was not supported.

HIGH FREQUENCY

$r > 4$

Geometry: $C_{\text{high}}(\sigma)$, $W_{\text{high}}(\sigma)$

selects two sampled candidates at $\sigma = 1.923, 1.088$

Condition	Noise level (sigma)	Final memorization
Before control	3.257, 5.315	65.23%
Candidate	1.088, 1.923	55.08%
After control	0.296, 0.585	66.41%

Result: only high-derived suppression was supported.

Interpretation

The intervention swaps the whole denoiser at the listed σ values; it does not swap Fourier components.

The high candidate is two sampled levels, not a continuously validated interval.